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A new crude oil price forecasting model based on variational mode decomposition

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ABSTRACT

Crude oil price prediction helps to get a better understanding of the global economic situation. Recently, variational mode decomposition (VMD) is introduced into the field of crude oil price forecasting. However, there is a lack of general selection rule for VMD-parameter and the widely adopted one-time decomposition strategy seems not suitable for practical application. Thus, an improved signal-energy based (ISE) rule is proposed as an improvement of the existing signal-energy based (SE) rule for the VMD-parameter selection. The moving-window strategy is put forward as a supplement for the decomposition strategy. Finally, a prediction model (VMD-LSTM-MW model) is built by combining the VMD, the long short-term memory (LSTM) network, and the moving-window strategy. The effectiveness of the ISE rule, the validity of the moving-window strategy, and the superiority of the VMD-LSTM-MW model are demonstrated by conducting monthly and daily crude oil price prediction experiments.

1. Introduction

Crude oil price plays an essential role in the global economy. It widely influences global economic activities, from economic policy-making to corporate profitability [1]. The decisive influence it has on global economic affairs has given rise to an increased research interest in the crude oil market [2]. Crude oil price forecasting is one of the most heated topics. However, the crude oil price is not only affected by the supply and demand [3] but also influenced by other factors like financial market and economic growth [4]. Due to the complex relationship between the crude oil market and the aforementioned factors, the time series of the crude oil price is non-linear and non-stationary [5]. Therefore, predicting crude oil prices accurately is a quite challenging task. Much research has been proposed to tackle it.

In the early days' researches, many traditional econometric models are proposed. The traditional econometric models could be divided into three types [6], namely, naive models, exponential smoothing models, and autoregressive family models, such as autoregressive integrated moving average (ARMIA) model [7], autoregressive conditional heteroskedasticity (ARCH) model [8], and generalized autoregressive conditional heteroskedasticity (GARCH) model [9]. Due to the great performance, the autoregressive family models are popular in modeling and forecasting the volatility of crude oil prices [6]. The traditional econometric models are very classic and lay a solid foundation for the development of crude oil price forecasting. However, the conventional models are considered not able to accurately describe the non-linear components of the crude oil price time series [6].

With the development of computing technologies, many modern methods are also developed. The main feature of them is the introduction of artificial intelligence. Equipped with artificial neural networks (ANN) [10–12], support vector machine (SVM) [13,14], visibility graph [15,16] or network forecasting methods [17], etc, the modern methods are capable to cope with the non-stationary features of the crude oil price time series and achieve promising performance [18]. Based on the type of model input, we divide the modern models into three categories: time-series models, structural models, and signal-decomposition models, which are shown in Table 1.

The time-series model uses only the crude oil price time series as its input. For example, in the recent researches, Khan et al. [19] proposed a prediction model combining the chicken swarm optimization algorithm and artificial neural network, which enhanced the performance of the bio-inspired neural network prediction models; Bristone et al. [20] utilized visibility graph to map the price time series on a network and employed K-core centrality to extract the non-linearity features of crude oil price dataset; just to name a few. However, the singularity of the input variables makes it difficult for the time-series model to possess adequate information for prediction.

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Table 1
Classification of the modern methods.

	Method	Feature
Modern methods	Time-series model	Utilize only the crude oil price time series itself as input.
	Structural model	Use more market information as inputs.
	Signal-decomposition model	Utilize the decomposed series of crude oil price time series as inputs.

The structural model is built to tackle the above problem. With more factors and variables introduced, the structural model achieves great performance. For example, in paper [21], with crude oil demand and supply, monthly GDP and CPI as factors, the decision tree model achieved better results; paper [22] suggested the important role the news texts could play in crude oil price forecasting; paper [23] even used Twitter sentiment of US foreign policy as input variable; just to name a few. However, the performance of the structural model is very sensitive to the quality of the factors while choosing appropriate factors is a challenging task, which requires professional economic knowledge and extensive research and investigation.

In order to feed the model enough information but avoid the complicated factor-choosing problem, the signal-decomposition model is proposed. Though it also utilizes only the crude oil price time series as input, the actual input of it is multiple. With signal decomposition techniques adopted, the time series is decomposed into several sub-series. The sub-series contains information at different frequency horizons, with some of them reflect fluctuations, and the others represent the long-term trends of the original series. The signal-decomposition model could be classified based on the decomposition methods [24], such as wavelet method [25–27], empirical mode decomposition (EMD) [28,29], ensemble empirical mode decomposition (EEMD) [30,31], seasonal adjust methods [32], and variational mode decomposition (VMD) [33], etc. However, the wavelet method has limitations, such as non-adaptive nature [34]. EMD and EEMD also suffer from their sensitivity to noise and sampling [35]. As a recently developed method, the effectiveness and the superiority of the VMD have been proven in other prediction areas, such as COVID-19 cases forecasting [36], wind speed prediction [37], streamflow forecasting [38], etc. Thus, it attracts lots of interest. Recently, the VMD method [35] is introduced into the crude oil price prediction.

Lahmiri [39] proposed a crude oil price forecasting model combining the generalized regression neural network (GRNN) and the VMD. It proved that the combination of VMD and GRUNN is better than the combination of EMD and GRUNN. Jianwei et al. [33] combined ARMIA and the VMD, achieving better performance than the EEMD based ARMIA model. Paper [18] proposed a method mixing the robust random vector functional link network (RVFLN) and the VMD. The proposed method beat several crude oil price forecasting traditional methods. Li et al. [5] combined the SVM and the VMD, obtaining promising results in monthly crude oil price prediction. Apart from crude oil price prediction, it is also adopted to analyze the systematic risk in the oil market [40]. However, most of the VMD-based crude oil price forecasting models utilized the one-time decomposition strategy (Details of it refer to Section 3.2). The one-time decomposition strategy bases on the assumption that future data is available during the decomposition process. Models adopted this strategy may not be suitable in practical cases [41]. Besides, there is a lack of general parameter-selection rules for the VMD [42]. In addition, a model combining the long short-term memory (LSTM) network and the VMD is popular in the area of wind speed forecasting [43,44] and is recently introduced into the area of non-ferrous metals price prediction [45].

Motived by the above reasons, we proposed our crude oil price prediction model by combining the VMD and the LSTM network. The contributions of this paper are written below:

1. We proposed an improved signal-energy based parameter-choosing rule for the VMD. Experiments in Section 4 indicate its superiority to the existing parameter-selection methods.
2. We proposed a novel strategy of decomposition, which is more practical compared to the one-time strategy. Experiments in Section 4 shows its superiority to the existing strategies.
3. A crude oil price prediction model is proposed by combining the VMD, the LSTM network, the proposed parameter-choosing rule, and the proposed decomposition strategy. Experiments in Section 4 demonstrates its superiority to the baseline models.

The remaining of the paper is structured as follows: Section 2 provides the basic knowledge of the VMD and the LSTM. In Section 3, the proposed parameter-choosing rule, the proposed decomposition strategy, and the proposed model are provided. In Section 4, we evaluate the proposed methods and model. Section 5 concludes the whole paper.

2. Methodology

In this section, two methods adopted in the proposed model will be described. In Section 2.1, we will introduce the VMD briefly. Then, the basic knowledge of the LSTM network will be provided in Section 2.2.

2.1. Variational mode decomposition

In 2013, Dragomiretskiy and Zosso proposed the VMD method [35]. As a non-recursive algorithm, the VMD concurrently decomposes the input signal into an ensemble of modes $\{u_k|k = 1, 2, \ldots, K\}$ and outputs their center frequencies $\omega_k (k = 1, 2, \ldots, K)$. K is a pre-defined parameter, determining the number of the modes extracted. Each mode u_k obtained fits the definition of the intrinsic mode function, which is an amplitude-modulated-frequency-modulated signal. To obtain the bandwidth of the mode, the following three steps should be executed [35]:

1. For each mode u_k, the Hilbert transform is utilized to generate its analytical signal and obtain the unilateral frequency spectrum.
2. The mode's frequency spectrum is shifted to baseband by mixing with an exponential tuned to the respective computed central frequency.
3. The bandwidth of u_k is then estimated through the H1 Gaussian smoothness of the demodulated signal.

After the estimation of the bandwidth, the VMD algorithm then becomes a constrained variational problem, written as [35]:

$$\min_{\{u_k\},\{\omega_k\}} \left\{ \sum_{k=1}^{K} \left\| \partial t \left[\left(\delta(t) + \frac{j}{\pi t} \right) * u_k(t) \right] e^{-j\omega_k t} \right\|_2^2 \right\}, \quad \text{s.t.} \sum_{k=1}^{K} u_k(t) = f(t), \tag{1}$$

where t represents the time script, $f(t)$ is the signal needed to be decomposed, δ implicates the Dirac distribution, u_k and ω_k are the

kth mode and its corresponding center frequency respectively, j is complex square root of -1, and $*$ denotes the convolution operator. $\sum_{k=1}^{K} u_k(t) = f(t)$ shows how the modes extracted, $u_k(k = 1, 2, \ldots, K)$, reproduce the original signal, $f(t)$.

In order to tackle the above problem more easily, the VMD adopts a quadratic penalty term and Lagrange multiplier (λ) to transform the constrained optimization problem to an unconstrained one. In this way, the augmented Lagrange $\mathfrak{L}$ is defined as [35]:

$$\begin{aligned}\mathfrak{L}(\{u_k\}, \{\omega_k\}) := \alpha \sum_{k=1}^{K} \left\| \partial t \left[\left(\delta(t) + \frac{j}{\pi t} \right) * u_k(t) \right] e^{-j\omega_k t} \right\|_2^2 \\ + \left\| f(t) - \sum_{k=1}^{K} u_k(t) \right\|_2^2 + \left\langle \lambda(t), f(t) - \sum_{k=1}^{K} u_k(t) \right\rangle,\end{aligned} \tag{2}$$

where α represents the balancing parameter of the data-fidelity constraint.

After that, a method called alternate direction method of multipliers (ADMM) [46] is utilized to find the solution of the above equation. The details of the ADMM optimization concept for VMD and the complete optimization of the VMD refer to paper [35].

Consider the whole VMD process as a function *VMD*(). Then, using the VMD to decomposed the input signal $f(t)$ into an ensemble of modes $\{u_k | k = 1, 2, \ldots, K\}$ could be written as [35]:

$$\{u_k(t) | k = 1, 2, \ldots, K\} = VMD(f(t), K), \tag{3}$$

where K is the pre-defined parameter, which determines the number of the modes extracted. However, in practical applications, the input signal is usually discrete. Hence, with n as the discrete time script, the above equation could be written as [35]:

$$\{u_k[n] | k = 1, 2, \ldots, K\} = VMD(f[n], K). \tag{4}$$

2.2. Long short-term memory network

The LSTM network, proposed by Hochreiter and Schmidhuber [47], is one of the most effective methods in the area of time-series forecasting. Compared to the artificial neural network (ANN), support vector machine (SVM), or other modern time series prediction methods, the LSTM network has higher prediction accuracy and lower variance [48]. Unlike the traditional neural networks, the LSTM network shows a satisfying performance when processing dynamic and non-linear time-series data, maintaining high prediction accuracy [45]. Therefore, it is widely adopted in many prediction applications, such as highway trajectory prediction [49], forecasting geo-sensory time-series [50], and even weather forecasting [51], etc. What is more, models combining the VMD and the LSTM network are popular in the area of wind speed forecasting [52] and are recently introduced into non-ferrous metals price prediction [45]. Thus, inspired by the previous researches, we adopt the LSTM network in our VMD based crude oil price forecasting model. Some basic knowledge of the LSTM network is provided below.

The LSTM network is made up of three parts, namely, an input layer, an output layer, and several recursive hidden layers between them [45]. In each hidden layer, there exist several memory modules. Inside the memory modules, there are some self-connected memory cells with three gates, namely, the input gate, the forgetting gate, and the output gate, controlling the information flow. It is the implementation of the memory cells and the nonlinear gating units that make the LSTM network capable to process long term and non-stationary sequences [53].

With the input vector $X = (x_1, x_2, \ldots, x_T)$, the hidden layer sequentially determines the activation values of the input gate I_t, the forgetting gate F_t, and the output gate O_t and calculates the cell state C_t of the memory cell, according to time $t = 1 \sim T$ [45]. The calculation process corresponding to time t is written as follows:

1. The memory cell reads in the input x_t and the previous hidden state h_{t-1}. Then, the forgetting gate determines which information should be abandoned using the equation below:

$$F_t = \sigma[W_F(h_{t-1}, x_t) + b_F]. \tag{5}$$

2. Then, the input gate I_t is updated and a new candidate vector $\tilde{C}_t$ of the cell state is generated according to the following equations:

$$I_t = \sigma[W_I(h_{t-1}, x_t) + b_I]. \tag{6}$$

$$\tilde{C}_t = \tanh[W_C(h_{t-1}, x_t) + b_C]. \tag{7}$$

3. After that, the cell state will be updated as follows (C_{t-1} is the previous cell state stored in the memory cell):

$$C_t = F_t \times C_{t-1} + I_t \times \tilde{C}_t. \tag{8}$$

4. Finally, the output gate O_t is calculated and the final output h_t is decided according to the following equations:

$$O_t = \sigma[W_O(h_{t-1}, x_t) + b_O]. \tag{9}$$

$$\tilde{h_t} = O_t \times \tanh(C_t). \tag{10}$$

In the above equations, $\sigma()$ represents the sigmoid function, W_F, W_I, W_C, W_O implicate the peephole connections, and b_F, b_I, b_C, b_O denote the input weight matrix.

3. The proposed methods and model

In this section, the proposed methods and model will be described. In Section 3.1, we will discuss the existing parameter-choosing rule of the VMD and propose an improved one basing on it. Then, in Section 3.2, the current decomposition strategies will be discussed and a novel strategy will be proposed. After that, in Section 3.3, the details of the proposed model will be provided.

3.1. Rules of choosing the pre-defined parameter K of the VMD

According to Eq. (4), it can be seen that the inputs of the VMD method are not only the original signal $f[n]$ but also a pre-defined parameter K. K determines how many modes will be extracted during the decomposition process. If too many modes extracted, it gives rise to accuracy reduction and unnecessary computing overheads [45]. However, if the number of modes is too small, the information contained in the modes is not enough to train a prediction model with high accuracy. Therefore, how to choose an appropriate value for K is quite important.

There is still a lack of general guidelines for the selection of the VMD-parameter [42]. Some new methods have been proposed in other areas to determine the number of the mode of the VMD. For example, Huang et al. [54] used the genetic algorithm to search the proper parameters and Li et al. [55] proposed to choose the number of the mode by calculating the effective Kurtosis index. However, the most common way adopted in the area of crude oil price forecasting to choose K is utilizing the EMD method to evaluate how many modes should be decomposed. Recently, in the area of non-ferrous metals price forecasting, a signal-energy based parameter-choosing rule [45] is proposed. The method proposed in [45] is very convenient to implement. What is more, it has been proven that the performance of the prediction model utilizing the K chose by the signal-energy based rule is better than that of the model adopting the K selected by the EMD.

The signal-energy based (SE) rule [45]: When the ratio of residual energy to the original signal energy, R_{res}, is small enough and there is no significantly downward trend, K can be determined [45]. The definition of R_{res} is given as follows:

$$R_{res} = \frac{1}{N}\sum_{n=1}^{N}\left|\frac{f[n]-\sum_{k=1}^{K}u_k[n]}{f[n]}\right| \times 100\%, \tag{11}$$

where $u_k[n]$ is the value of nth time script of kth mode of the extracted ensemble $\{u_k[n]|k = 1, 2, \ldots, K; n = 1, 2, \ldots, N\}$, $f[n]$ is the original signal, and N is the length of $f[n]$. R_{res} is actually the mean absolute percentage error (MAPE) between the reconstructed signal, $\sum_{k=1}^{K} u_k[n]$, and the original signal, $f[n]$.

Though the effectiveness of the above SE rule has been proven, it does not reflect the actual energy of the signal. Thus, we propose a new rule basing on it.

The proposed rule: the improved signal-energy based (ISE) rule When the ratio of residual energy to the original signal energy, E_{res}, it is small enough and there is no significant downward trend, K can be determined. Though the description of the ISE rule is the same as the SE rule, the definition of E_{res} is different from R_{res}. Inspired by the definition of the energy of finite discrete-time signal, E_{res} is defined as follows:

$$E_{res} = \frac{\sum_{n=1}^{N}\left|f[n]-\sum_{k=1}^{K}u_k[n]\right|^2}{\sum_{n=1}^{N} f[n]^2} \times 100\%, \tag{12}$$

In this way, E_{res} reflects the ratio of the actual residual energy to the actual energy of the original signal. If the reduction between the two values is less than 1%, we considered there is no significant downward trend between these two values. The proposed rule will be evaluated in Section 4.

3.2. Decomposition strategies

Among the current research of time series forecasting using the VMD, there are two main strategies. One is called the one-time strategy, the other is named as the real-time strategy. Many existing VMD or EMD based researches in crude oil price forecasting, such as paper [5,18,29,33,39,56], just to name a few, adopt the one-time strategy.

One-time strategy The main feature of this strategy is that the whole raw data of the crude oil price time series is decomposed before being divided into the training part and the assessment part. Therefore, during the training and the prediction process, the decomposition is only applied once, which is why this strategy is called the "one-time strategy" [41]. To illustrate this strategy more clearly, the pseudo-code of model adopting this strategy is provided in algorithm 1.

Algorithm 1 One-time strategy

//See the description of symbols at the end of Section 3.2.
//Initialization part
(1) Decompose the raw data series $f[n]$ using VMD. According to Eq. (4), the process could be written as:

$$\{u_k[n]|k = 1, 2, \ldots, K; n = 1, 2, \ldots, N_T + N_A\} = VMD(f[n], K).$$

(2) Train the model with the mode data of the training part, which is $\{u_k[n]|k = 1, 2, \ldots, K; n = 1, 2, \ldots, N_T\}$, as inputs.
//Prediction part
//Prediction period is from day $N_T + 1$ to day $N_T + N_A$.
(3) $count \Leftarrow 0$. // Temporary variable for counting
repeat
(4) $count \Leftarrow count + 1$.
(5) Predict the $count$ th day's data of the prediction period using the well-trained model with the mode data of day $count - 1$ as inputs, written as:

$$P_{out} = Model(\{u_k[n]|k = 1, 2, \ldots, K; n = N_T + count - 1\}).$$

until $count == N_A$.

However, if the one-time strategy adopted, the resulting components utilized to predict the future value of a particular moment are extracted using information not only from historical values but also from future values [57]. Since the future values are not available in practical prediction, these models may not suitable for real forecasting [41,57]. To tackle this problem, an adaptive forecasting model is proposed [58]. The decomposition strategy used in this research is called "real-time strategy".

Real-time strategy The main feature of this strategy is that only the historical data of the raw data of the crude oil price time series would be decomposed. For example, if we want to predict the D th day's data, original signal $f[n](n = 1, 2, \ldots, D-1)$ will be decomposed. Thus, the decomposition process happens in every prediction iteration. The pseudo-code of the model adopting this strategy is provided in algorithm 2. As it is shown in algorithm 2, we could choose when to update the historical data and retrain the model. If the model is retrained every iteration, the model could possess new information once new data arrives. If the model is retrained when the prediction accuracy drops with the evolution of time, more computing resources would be saved.

Algorithm 2 Real-time strategy

//See the description of symbols at the end of Section 3.2.
//Initialization part
(1) $count \Leftarrow 0$. // Temporary variable for counting
(2) Obtain historical data:

$$H[n] \Leftarrow f[n](n = 1, 2, \ldots, N_T + count)$$

(3) Decompose the historical data series $H[n]$ using VMD. According to Eq. (4), the process could be written as:

$$\{u_k[n]|k = 1, 2, \ldots, K; n = 1, 2, \ldots, N_T + count\} = VMD(H[n], K).$$

(4) Train the model with the mode data extracted in step 3, which is $\{u_k[n]|k = 1, 2, \ldots, K; n = 1, 2, \ldots, N_T + count\}$, as inputs.
//Prediction part
//Prediction period is from day $N_T + 1$ to day $N_T + N_A$.
repeat
(5) $count \Leftarrow count + 1$.
(6) Obtain the prediction input through decomposition:

$$\{u_k[n]|k = 1, 2, \ldots, K; n = 1, 2, \ldots, N_T + count - 1\} = VMD(f[n](n = 1, 2, \ldots, N_T + count - 1), K).$$

(7) The prediction input P_{in} of the $count$ th day's data of the prediction period is the mode data of day $count - 1$, which is:

$$P_{in} \Leftarrow \{u_k[n]|k = 1, 2, \ldots, K; n = N_T + count - 1\}.$$

(8) Predict the $count$ th day's data of the prediction period using the well-trained model with P_{in} as input, written as:

$$P_{out} = Model(P_{in}).$$

//Update part

if needed **then**
//If needed, update the historical data and retrain the model.
Pause the loop.
Update the historical data:

$$H[n] \Leftarrow f[n](n = 1, 2, \ldots, N_T + count).$$

Execute step 3 to step 4.
Restart the loop.
end if
until $count == N_A$.

Inspired by the real-time strategy, we proposed a novel decomposition strategy, called "moving-window strategy"

The proposed strategy: moving-window strategy The whole process of this strategy is quite similar to the real-time strategy. The only difference between them is how the prediction input is generated. In this strategy, only a certain length of data known for the day D, which is $f[n](n = D - Size_w, \ldots, D - 1)$ would be used to generate the prediction input. $Size_w$ is a constant, called the window size. That is to say, unlike the real-time strategy, the length of data used for prediction-input generation is fixed and would not extend during the forecasting. If the $Size_w$ is changed to a variable and set as the whole length of the historical data, the moving-window strategy would become the real-time strategy. The pseudo-code of the model adopting this strategy is provided in algorithm 3. It can be seen that, the only difference between algorithm 2 and 3 is step 6.

Algorithm 3 The proposed strategy: moving-window strategy

//See the description of symbols at the end of Section 3.2.
//Initialization part
//$Size_w$ is the window size.
(1) $count \Leftarrow 0$. // Temporary variable for counting
(2) Obtain historical data:

$$H[n] \Leftarrow f[n](n = 1, 2, \ldots, N_T + count)$$

(3) Decompose the historical data series $H[n]$ using VMD. According to (4), the process could be written as:

$$\{u_k[n]|k = 1, 2, \ldots, K; n = 1, 2, \ldots, N_T + count\} = VMD(H[n], K).$$

(4) Train the model with the mode data extracted in step 3, which is $\{u_k[n]|k = 1, 2, \ldots, K; n = 1, 2, \ldots, N_T + count\}$, as inputs.
//Prediction part
//Prediction period is from day $N_T + 1$ to day $N_T + N_A$.
repeat
(5) $count \Leftarrow count + 1$.
(6) Obtain the prediction input through decomposition:

$$\{u_k[n]|k = 1, 2, \ldots, K; n = N_T + count - Size_w, \ldots, N_T + count - 1\} = VMD(f[n](n = N_T + count - Size_w, \ldots, N_T + count - 1), K).$$

(7) The prediction input P_{in} of the *count* th day's data of the prediction period is the mode data of day $count - 1$, which is:

$$P_{in} \Leftarrow \{u_k[n]|k = 1, 2, \ldots, K; n = N_T + count - 1\}.$$

(8) Predict the *count* th day's data of the prediction period using the well-trained model with P_{in} as input, written as:

$$P_{out} = Model(P_{in}).$$

//Update part
if needed **then**
//If needed, update the historical data and retrain the model.
Pause the loop.
Update the historical data:

$$H[n] \Leftarrow f[n](n = 1, 2, \ldots, N_T + count).$$

Execute step 3 to step 4.
Restart the loop.
end if
until $count == N_A$.

To be noticed, in all the pseudo-codes provided in this part, which are algorithms 1, 2, and 3, $f[n]$ represents the raw data series, N_T is the length of training part of $f[n]$, N_A is the length of assessment part of $f[n]$, $Model()$ implicates the well-trained model, $u_k[n]$ represents the modes extracted, K is the number of modes extracted, and P_{out} is the prediction output. Thus, the length of $f[n]$ is $N_T + N_A$. We also assume that models in the above algorithms are one-day ahead models and only use the data of the last trading day to predict the oil price on the next trading day.

3.3. The proposed crude oil price forecasting model

We proposed our prediction model, the VMD-LSTM-MW model, by combining the LSTM network and the VMD, with the moving-window strategy adopted. In the VMD-LSTM-MW model, the VMD adopting moving-window strategy is utilized to generate the training inputs and the prediction inputs of the LSTM network. The schematic description of the VMD-LSTM-MW model is demonstrated in Fig. 1. The proposed model should be implemented following the procedure below:

(i) Firstly, we obtain historical data series.
(ii) In the training part, we generate the parameter of the VMD using the ISE rule proposed in Section 3.1. Then, with the parameter produced by the ISE rule, we calculate the training input according to Algorithm 3 step (1) to (2). After that, train the LSTM network sufficiently with the training input.
(iii) In the prediction part, we generate the prediction input and predict the result using the well-trained LSTM network according to Algorithm 3 step (3) to (4).

More details about how to use the moving-window strategy to construct the VMD-LSTM-MW model refers to algorithm 3. The VMD-LSTM-MW model is a one-day-ahead model.

4. Experiment

In this section, we assess the performance of the proposed VMD-LSTM-MW model. The effectiveness of the proposed rule and the proposed strategy are also evaluated. Monthly and daily crude oil price predictions are conducted in Sections 4.1 and 4.2, respectively. Multi-step-ahead prediction is also conducted in Section 4.1 to assess the robustness of our model.

4.1. Experiment 1: monthly crude oil price prediction

4.1.1. Settings of experiment 1

Data Set 1 We use the same data set as in paper [5], which is the crude oil monthly price of the WTI market. The duration of the data mentioned above is from January 1994 to July 2018. The graphical version of data set 1 is provided in Fig. 2. In the crude oil monthly price series, the first 80% of the series is regarded as the historical data, while the remaining 20% of the series is

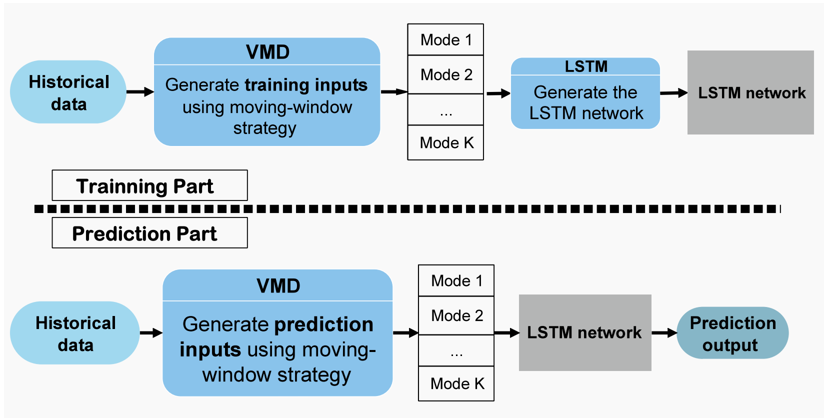

Fig. 1. The schematic description of the VMD-LSTM-MW model.

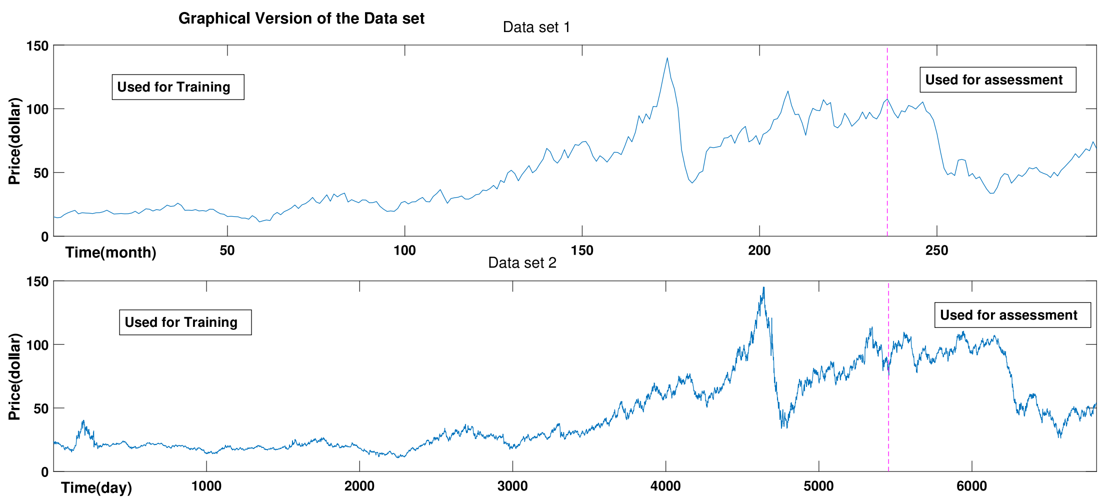

Fig. 2. The graphical version of data sets.

used for assessment according to paper [5]. Please refer to [5] for more information about the dataset and the adopted experiment method.

$$MAPE = \frac{1}{T}\sum_{i=1}^{T}\left|\frac{(Value_{actual}(i) - Value_{predicted}(i))}{Value_{actual}(i)}\right| \times 100\%,$$
$$RMSE = \sqrt[2]{\frac{\sum_{i=1}^{T}(Value_{actual}(i) - Value_{predicted}(i))^2}{T}}. \tag{13}$$

Evaluation Metrics According to the Ref. [5], we select the mean absolute percentage error (MAPE), and the root mean of squared errors (RMSE) as evaluation metrics. Their definitions are listed in Eq. (13).

Settings of the LSTM network Referring to the recommendations in paper [45] and basing on our experience, we set the parameters of the LSTM network as follows: the number of the hidden layers is 5; the batch size is set as 10; learning rate is 0.005; the time step is set as 5, namely using the data from the first 5 months to predict the monthly price of next month. More details about the parameters setting of the LSTM network please refer to [45]. Another method to set the parameters of the LSTM network please see Ref. [59].

Baseline Models According to paper [5], the ARMIA model, the genetic algorithm optimized SVM (GA-SVM) model, model combining the EEMD and the GA-BPNN (EEMD-GA-SVM), and the model combining the VMD and the GA-SVM (VMD–GA-SVM) are chosen to be the baseline models. The details of the above models please see Ref. [5].

Table 2
The ratio of residual energy corresponding to different K under different parameter-choosing rules (monthly crude oil price prediction).

K	Chosen by the EEMD	R_{res} (%)	Chosen by the SE [45]	E_{res} (%)	Chosen by the ISE (**Ours**)
3		3.25E−04		1.31E−05	
4		2.56E−04		7.71E−06	
5		1.81E−04		1.96E−06	
6		1.79E−04		1.33E−06	
7		1.78E−04		1.21E−06	
8	✔	**1.74E−04**	✔	1.19E−06	
9		1.76E−04		1.10E−06	
10		**1.77E−04**	✔	**1.04E−06**	✔
11		1.77E−04		1.03E−06	
12		1.77E−04		1.02E−06	

4.1.2. Evaluation of the proposed parameter-choosing rule

In this part, we assess the effectiveness of the proposed VMD-parameter-selection rule. The ratio of residual energy corresponding to different K under different parameter-choosing rules is provided in Table 2. In paper [5], K is set as 8 using the EEMD method. According to the SE rule, as it is shown in Table 2, no significant downward trend is witnessed after $K = 10$. Thus, K should be set as 10. However, R_{res} has a minimum value at $K = 8$.

Table 3
Performance of the VMD-LSTM-MW models corresponding to different K (monthly crude oil price prediction).

K	MAPE (%)	RMSE
8 (chosen by the EEMD and the SE)	3.09	2.47
9	3.28	2.38
10 (chosen by the ISE (**Ours**) and the SE)	**2.03**	**1.30**
11	2.31	1.49
12	3.86	3.35

Table 4
Performance of the VMD-LSTM models corresponding to different decomposition strategies (monthly crude oil price prediction).

Strategies	MAPE (%)	RMSE
one-time	4.59	3.61
real-time	6.30	4.82
moving-window ($Size_w = 2$)	**2.03**	**1.30**
moving-window ($Size_w = 3$)	5.53	3.39
moving-window ($Size_w = 4$)	2.91	3.08
moving-window ($Size_w = 5$)	4.51	2.83
moving-window ($Size_w = 10$)	10.96	8.68
moving-window ($Size_w = 20$)	10.60	9.24
moving-window ($Size_w = 30$)	7.82	6.79
moving-window ($Size_w = 40$)	7.40	7.14
moving-window ($Size_w = 50$)	8.76	8.19
moving-window ($Size_w = 100$)	5.84	4.54

Therefore, $K = 8$ and $K = 10$ are considered as the choices of the SE rule. When it comes to the proposed ISE rule, the reduction of E_{res} after $K = 10$ is less than 1%, thus, $K = 10$ is selected. The performance of the VMD-LSTM-MW models corresponding to the selected K is provided in Table 3. Models with $K = 9, 11, 12$ are also constructed respectively, to demonstrate the effectiveness of the ISE rule. In this part, the window-size of the VMD-LSTM-MW models is set as 2.

It can be seen that, in Table 3, both MAPE and RMSE obtain their minimum value when $K = 10$. The MAPE of the model with $K = 10$ is 34.45%, 38.23%, 12.06%, and 47.38%, better than that of the models with $K = 8, 9, 11$, and 12. The RMSE of the model with $K = 10$ is reduced by 47.39%, 45.50%, 12.74%, and 61.22% compared to that of the models with $K = 8, 9, 11$, and 12. The experiment results indicate that the VMD-parameter selected by the ISE rule is better than the VMD-parameter selected by the EEMD method.

Though $K = 10$ is also selected by the SE rule, it also chooses $K = 8$ with which the model does not perform quite well. In other words, the result generated by the SE rule is not very conclusive as we still need to test more than one mode to select the best. Thus, using the proposed ISE rule instead of the EEMD method and the SE rule to set the VMD-parameter is more appropriate in this experiment. The effectiveness of the proposed ISE is proven.

4.1.3. Evaluation of the proposed decomposition strategy

In this part, we assess the effectiveness of the decomposition strategy. According to Section 4.1.2, K is set as 10. VMD-LSTM models using one-time strategy and real-time strategy are constructed respectively. The performance of the VMD-LSTM models corresponding to different decomposition strategies are provided in Table 4. In the moving-window strategy, we set the tested window size as follows: $Size_w = 2, 3, 4, 5, 10, 20, 30, 40, 50, 100$.

It can be seen that, in Table 3, both MAPE and RMSE obtain their minimum value when moving-window strategy adopted with $Size_w = 2$. When the $Size_w$ of the moving-window strategy is below 5, the performance of the moving-window strategy models is better than that of the one-time strategy model and the real-time strategy model. However, when the $Size_w$ becomes bigger, the prediction accuracy of the moving-window strategy is not quite satisfying. The experiment results show the potential of the moving-window strategy. If the $Size_w$ is set appropriately, the moving-window strategy model would be superior to models adopting other strategies. Thus, how to set the $Size_w$ more appropriately would be our future study.

Table 5
One-day-ahead monthly crude oil price forecasting performance of different models.

Models	MAPE (%)	RMSE
VMD-LSTM-MW	**2.03**	**1.30**
VMD-GA-SVM[a]	3.14	1.89
EEMD-GA-SVM[a]	3.39	2.40
GA-SVM[a]	7.26	5.42
ARIMA[a]	6.19	4.57

[a]The statistics of the evaluation metrics of the model quote from paper [5].

Table 6
Two-day-ahead monthly crude oil price forecasting performance of different models.

Models	MAPE (%)	RMSE
VMD-LSTM-MW	**3.74**	**2.32**
VMD-GA-SVM[a]	4.31	2.62
EEMD-GA-SVM[a]	4.97	3.50
GA-SVM[a]	11.42	8.34
ARIMA[a]	10.40	7.80

[a]The statistics of the evaluation metrics of the model quote from paper [5].

4.1.4. Evaluation of the proposed VMD-LSTM-MW model

In this part, we evaluate the monthly crude oil price prediction performance of the proposed VMD-LSTM-MW model, which is the model combining the VMD, the LSTM and the moving-window strategy. Some state-of-the-art models mentioned in Section 4.1.1 are used as comparisons. According to paper [5], we conduct multi-step-ahead forecasting. The one-day-ahead, two-day-ahead, and three-day-ahead prediction performance of different models are provided in Tables 5, 6, and 7, respectively.

It can be seen that, in Table 5, the VMD-LSTM-MW model shows great performance in one-day-ahead monthly crude oil price forecasting. It achieves the best prediction accuracy among the models. The MAPE of the VMD-LSTM-MW model is 35.39%, 40.16%, 72.06%, and 67.23% better than that of the VMD-GA-SVM model, the EEMD-GA-SVM model, the GA-SVM model, and the ARIMA model, respectively. The RMSE of the VMD-LSTM-MW model is reduced by 31.45%, 45.89%, 76.05%, and 71.57% compared to that of the VMD-GA-SVM model, the EEMD-GA-SVM model, the GA-SVM model, and the ARIMA model, respectively.

When it comes to two-day-ahead monthly crude oil price forecasting, as it is shown in Table 6, the VMD-LSTM-MW model maintains its superiority. It achieves the best prediction performance among the models. The MAPE of the VMD-LSTM-MW model is 13.32%, 24.83%, 67.29%, and 64.08% better than that of the VMD-GA-SVM model, the EEMD-GA-SVM model, the GA-SVM model, and the ARIMA model, respectively. The RMSE of the VMD-LSTM-MW model is reduced by 11.60%, 33.78%, 72.21%, and 70.28% compared to that of the VMD-GA-SVM model, the EEMD-GA-SVM model, the GA-SVM model, and the ARIMA model, respectively.

However, the performance accuracy of the VMD-LSTM-MW model drops significantly in three-day-ahead prediction. The MAPE and RMSE of the VMD-LSTM-MW model are 9.22 and 14.30, which taking the third and the last place among the models, respectively. RMSE could magnify the effect of outliers and show the deviation of predicted values. The high value of the RMSE indicates a high number of outliers or a high deviation of some predicted values. Thus, we could deduce that the

Table 7
Three-day-ahead monthly crude oil price forecasting performance of different models.

Models	MAPE (%)	RMSE
VMD-LSTM-MW	9.22	14.30
VMD-GA-SVM[a]	**6.06**	**3.82**
EEMD-GA-SVM[a]	6.91	5.55
GA-SVM[a]	16.23	12.01
ARIMA[a]	14.87	11.15

[a]The statistics of the evaluation metrics of the model quote from paper [5].

significant decrease of the prediction accuracy of the VMD-LSTM-MW model is that it outputs quite a lot outliers, or some of its predicted values have a large deviation, in three-day-ahead prediction. It seems that the proposed model is not suitable for multi-step-ahead forecasting.

4.2. Experiment 2: daily crude oil price prediction

The multi-step-ahead forecasting performance of the VMD-LSTM-MW model is not quite satisfying. Thus, in this subsection, we conduct daily crude oil price prediction to evaluate the performance of the proposed model when applied to a different time horizon forecasting. More baseline models are introduced for assessment.

4.2.1. Settings of experiment 2

Data set 2 We use the same data set as in paper [18], which is the closing price of the crude oil daily price of the WTI market. The duration of the data mentioned above is from 2 January 1990 to 30 December 2016. The graphical version of data set 2 is provided in Fig. 2. In the crude oil monthly price series, the first 80% of the series is regarded as the historical data, while the remaining 20% of the series is used for assessment according to paper [18]. Please refer to [18] for more information about the dataset and the adopted experiment method.

Evaluation metrics According to the Ref. [18], we select the mean absolute percentage error (MAPE) as evaluation metric. Its definition is listed in Eq. (13).

Settings of the LSTM network Referring to the recommendations in paper [45] and basing on our experience, we set the parameters of the LSTM network as follows: the number of the hidden layers is 5; the batch size is set as 50; learning rate is 0.005; the time step is set as 30, namely using the data from the first 30 trading days to predict the closing price of next trading day. More details about the parameters setting of the LSTM network please refer to [45]. Another method to set the parameters of the LSTM network please see Ref. [59].

Baseline models According to paper [18], the ARMIA model, the random vector functional link network (RVFLN) model, the conventional RVFLN (con-RVFLN) model, the BPNN model, the extreme learning machine (ELM) model, the model combining the RVFLN and the VMD (VMD–RVFLN), the model combining the RVFLN and EMD (EMD–RVFLN), the model combining the ELM and the VMD (VMD–ELM), the model combining the ELM and the EMD (EMD–ELM), the least square support vector regression (LSSVM) model, the adaptive neuro-fuzzy inference system (ANFIS) model, the interval type-2 fuzzy neural network (IT2FNN) model, and the recurrent neural network (RNN) model are chosen to be the baseline models. The details of the above models please see Ref. [18].

Table 8
The ratio of residual energy corresponding to different K under different parameter-choosing rules (daily crude oil price prediction).

K	R_{res} (%)	Chosen by the SE [45]	E_{res} (%)	Chosen by the ISE (**Ours**)
3	6.53E−03		8.11E−07	
4	2.93E−03		1.44E−07	
5	2.43E−03		9.59E−08	
6	1.70E−03		3.48E−08	
7	1.32E−03		1.52E−08	
8	1.28E−03		1.34E−08	
9	1.14E−03		6.66E−09	
10	**1.10E−03**	✓	4.70E−09	
11	1.10E−03		4.20E−09	
12	1.09E−03		**3.90E−09**	✓
13	1.09E−03		3.90E−09	

Table 9
Performance of the VMD-LSTM-MW models corresponding to different K (daily crude oil price prediction).

K	MAPE (%)
9	0.46
10 (chosen by the SE)	0.62
12 (chosen by the ISE (**Ours**))	**0.45**

4.2.2. Evaluation of the proposed parameter-choosing rule

In this part, we assess the effectiveness of the proposed VMD-parameter-selection rule. The ratio of residual energy corresponding to different K under different parameter-choosing rules is provided in Table 8. In paper [18], K is set as 16 using the EEMD method. However, when $K = 16$ adopted in the VMD method, the number of the modes extracted is less than 16. It means that when EEMD is adopted, the selected K may be too large for the VMD. According to the SE rule, as it is shown in Table 8, the reduction of R_{res} after $K = 10$ is less than 1%, thus, $K = 10$ is selected. When it comes to the proposed ISE rule, there is no significant downward trend of E_{res} after $K = 12$, thus, $K = 12$ is selected by the ISE rule. The performance of the VMD-LSTM-MW models corresponding to the selected K is provided in Table 9. The model with the randomly selected $K = 9$ is also constructed. In this part, the window-size of the VMD-LSTM-MW models is set as 2.

It can be seen that, in Table 9, the MAPE obtain its minimum value when $K = 12$. The MAPE of the model with $K = 12$ is 2.17% and 27.42% better than that of the models with $K = 9$ and 10. The experiment results indicate that the VMD-parameter selected by the ISE rule is better than the VMD-parameter selected by the SE method. The performance of the model with the parameter selected by the SE rule is even worse than that of the model with a randomly chosen parameter. Thus, using the ISE rule instead of the EEMD method and the SE rule to set the VMD-parameter is more appropriate in this experiment. The effectiveness of the ISE is proven.

4.2.3. Evaluation of the proposed decomposition strategy

In this part, we assess the effectiveness of the decomposition strategy. According to Section 4.2.2, K is set as 12. VMD-LSTM models using one-time strategy and real-time strategy are built respectively. The performance of the VMD-LSTM models corresponding to different decomposition strategies are provided in Table 10. In the moving-window strategy, the tested window size is set as follows: $Size_w = 2, 3, 4, 5, 10, 20, 30, 40, 50, 100$.

It can be seen that, in Table 3, MAPE obtains its minimum value when moving-window strategy adopted with $Size_w = 2$. Apart from $Size_w = 2$, when the $Size_w$ of the moving-window strategy is 4, 50 and 100, the performance of the moving-window strategy models is also better than that of the one-time strategy

Table 10
Performance of the VMD-LSTM models corresponding to different decomposition strategies (daily crude oil price prediction).

Strategies	MAPE (%)
one-time	1.12
real-time	1.15
moving-window ($Size_{window} = 2$)	**0.46**
moving-window ($Size_{window} = 3$)	3.84
moving-window ($Size_{window} = 4$)	0.59
moving-window ($Size_{window} = 5$)	3.43
moving-window ($Size_{window} = 10$)	1.67
moving-window ($Size_{window} = 20$)	1.53
moving-window ($Size_{window} = 30$)	1.31
moving-window ($Size_{window} = 40$)	1.21
moving-window ($Size_{window} = 50$)	1.04
moving-window ($Size_{window} = 100$)	0.78

Table 11
Daily crude oil price Forecasting performance of different models.

Models	MAPE (%)
VMD-LSTM-MW	**0.46**
VMD–RVFLN[a]	1.35
EMD–RVFLN[a]	1.62
VMD–ELM[a]	1.95
EMD–ELM[a]	2.56
RVFLN[a]	2.11
con-RVFLN[a]	2.25
ELM[a]	2.63
BPNN[a]	4.89
ARIMA[a]	4.17
LSSVR[a]	3.83
ANFIS[a]	6.02
IT2FNN[a]	5.78
RNN[a]	4.06

[a]The statistics of the evaluation metrics of the model quote from paper [18].

model and the real-time strategy model. However, when the $Size_w$ is 3 and 5, the prediction accuracy of the moving-window strategy is not quite satisfying. The experiment results above show the potential of the moving-window strategy. If the $Size_w$ is set appropriately, the moving-window strategy model would be superior to models adopting other strategies. Thus, how to set the $Size_w$ more appropriately would be our future study.

4.2.4. Evaluation of the proposed VMD-LSTM-MW model

In this part, we evaluate the daily crude oil price prediction performance of the proposed VMD-LSTM-MW model, which is the model combining the VMD, the LSTM, and the moving-window strategy. The provided VMD-LSTM-MW model is a one-day-ahead model. Some state-of-the-art models mentioned in Section 4.2.1 are used as comparisons. The prediction performance of different models is provided in Table 11.

It can be seen that, in Table 11, the VMD-LSTM-MW model shows great performance in daily crude oil price forecasting. It achieves the best prediction accuracy among the models. Compared with other models, the advantage of the VMD-LSTM-MW model is very large. The MAPE of the VMD-LSTM-MW model is 66.03%, 71.55%, 76.40%, 82.06%, 78.26%, 79.54%, 82.50%, 90.61%, 88.98%, 87.99%, 92.36%, 92.05%, and 88.69% better than that of the VMD–RVFLN model, the EMD–RVFLN model, the VMD–ELM model, the EMD–ELM model, the RVFLN model, the con-RVFLN model, the ELM model, the BPNN model, the ARIMA model, the LSSVR model, the ANFIS model, the IT2FNN model, and the RNN model, respectively.

4.3. Discussion

We discuss the findings of the experiments below:

1. According to the experimental results in Sections 4.1.1 and 4.2.1, models using the parameter chosen by the proposed ISE rule achieve better prediction accuracy, compared to the existing SE rule and the EEMD method. Also, compared to the SE rule, the result generated by the proposed ISE rule is more conclusive as more than one mode still need to be tested before the decomposition if the SE rule is adopted. Thus, the proposed ISE rule is superior to the existing rules. What is more, inspired by the energy of the discrete-time signal, its definition of the residual energy could reflect the actual energy of the residual signal of the decomposition.
2. The experimental results in Sections 4.1.2 and 4.2.2 suggests the effectiveness of the proposed moving-window strategy. Among the listed window size, models with $Size_w = 2$ achieve the best performance in both the two cases. The moving-window strategy models with $Size_w = 2$ are better than the one-time and the real-time strategy in both monthly and daily crude oil price forecasting. Though the performance of the model adopting the moving-window strategy is affected by the selected window size, its potential shown in the experiments should be recognized. How to set the $Size_w$ appropriately would be our future study.
3. The superiority of the proposed one-day-ahead VMD-LSTM-MW model is demonstrated in experiments in Sections 4.1.3 and 4.2.3. The proposed model is the best in both the daily and monthly crude oil price predictions. The forecasting accuracy of the VMD-LSTM-MW model drops in three-step-ahead monthly prediction. It indicates that the proposed model still needs future improvement for multi-step-ahead prediction. However, as a one-day-ahead forecasting model, its effectiveness is already proven by the above experiment. The robustness of the VMD-LSTM-MW model is also demonstrated by the conducted different time horizons forecasting. What is more, the VMD-LSTM-MW utilizes the moving-window strategy, but other baseline models use the one-time strategy. As we mentioned before, the one-time strategy may not be suitable for practical prediction [41,57]. Thus, compared to the baseline models, the proposed model is not only more accuracy-promising but also more practical.

5. Conclusions and future work

In this paper, three new things are proposed, namely, the ISE rule, the moving-window strategy, and the VMD-LSTM-MW model. Firstly, the ISE rule is proposed as an improvement of the SE rule, developed for the VMD-parameter selection. Compared to the SE rule, the definition of the residual energy of the ISE rule is more realistic. The effectiveness of the ISE rule is proven by experiments. Secondly, the moving-window strategy is put forward as a supplement for the decomposition strategy. Compared to the existing strategies, the model adopting this strategy could achieve better performance. The effectiveness of the moving-window strategy is proven by experiments. Finally, by combining the LSTM network, the VMD, and the moving-window strategy, the VMD-LSTM-MW model is proposed. The proposed model is superior to the baseline models in both monthly and daily crude oil price forecasting. In the future, we will improve the multi-step-ahead prediction capability of the VMD-LSTM-MW model and propose more appropriate methods to set the window size of the moving-window strategy.

CRediT authorship contribution statement

Yusheng Huang: Conceptualization, Methodology, Software, Writing - original draft, Visualization, Data curation. **Yong Deng:** Validation, Writing - review & editing, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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